

Manohar Kokkalakonda

AI/ML Engineer

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Summary

AI/ML Engineer with 5+ years of experience building production-grade machine learning, Generative AI, and RAG-based systems across healthcare, fintech, and payment technology domains. Skilled in Python, SQL, TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, LangChain, FastAPI, MLflow, Docker, Kubernetes, AWS, and Azure. Strong background in payment fraud detection, transaction risk scoring, merchant risk assessment, anomaly detection, NLP, real-time data pipelines, model governance, and cloud-native MLOps.

Technical Skills

- Programming Languages:** Python, SQL
- Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Pandas, NumPy, LSTM, Hugging Face Transformers, spaCy
- Generative AI & LLMs:** OpenAI API, Azure OpenAI, LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, FAISS, Pinecone, ChromaDB, LangSmith, RAGAS, Agentic AI Workflows
- Fintech, Fraud & Risk AI:** Payment Fraud Detection, Transaction Risk Scoring, Merchant Risk Assessment, Payment Success Prediction, KYC Automation, Dispute Categorization, Anomaly Detection, Real-Time Decisioning
- MLOps & Deployment:** MLflow, Docker, Kubernetes, FastAPI, Flask, Git, GitHub Actions, Azure DevOps, CI/CD, Model Registry, Model Monitoring, Model Versioning, Data Drift Monitoring
- Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker, Glue, Athena), Azure (Azure Machine Learning, Azure OpenAI, Azure AI Search, Data Factory, Data Lake Storage, Event Hub)
- Data Engineering:** Apache Spark, Apache Airflow, Apache Kafka, Hive, ETL Pipelines, Real-Time Streaming, Feature Engineering
- Databases & Warehousing:** MySQL, PostgreSQL, MongoDB, Redis, Snowflake, Google BigQuery
- Visualization & Explainability:** Power BI, Tableau, SHAP, LIME, Grafana
- Security & Compliance:** HIPAA, GDPR, PII Protection, PHI Protection, SOC2, Audit Logging, AI Governance

Professional Experience

- AI/ML Engineer, Cigna Healthcare

01/2024 – Present | Remote, USA

 - Designed and deployed machine learning models for healthcare claims prediction, fraud detection, anomaly detection, and clinical risk analytics using Python, TensorFlow, and Scikit-learn, improving fraud detection precision by 17% while reducing false claim payouts by 22%.
 - Built scalable streaming data pipelines using Azure Event Hub, Apache Kafka, Apache Spark, Apache Airflow, and AWS Glue to process over 10 million healthcare records daily, enabling real-time analytics, fraud detection, and predictive decision-making.
 - Developed production-grade AI services using FastAPI and Flask, exposing ML models through REST APIs and integrating them with enterprise healthcare applications while maintaining sub-300 ms inference latency.
 - Implemented end-to-end MLOps pipelines using Azure Machine Learning, Docker, Kubernetes, MLflow, Azure DevOps, and Grafana for automated model training, deployment, monitoring, model versioning, and drift detection, increasing deployment frequency by 35%.
 - Developed enterprise Retrieval-Augmented Generation (RAG) applications using Azure OpenAI, LangChain, LangGraph, LlamaIndex, Azure AI Search, and FAISS to automate prior authorization workflows, clinical document retrieval, and knowledge search.
 - Built LLM evaluation pipelines using LangSmith and RAGAS, implementing AI guardrails, prompt evaluation, human-in-the-loop review workflows, and audit logging to improve response quality while supporting HIPAA and enterprise AI governance requirements.
- ML Engineer, Razorpay

01/2020 – 07/2022 | Hyderabad, India

 - Developed and deployed machine learning models for payment fraud detection, transaction risk scoring, payment success prediction, and merchant risk assessment using Python, Scikit-learn, XGBoost, and LightGBM, improving fraud detection precision by 18% and reducing false-positive alerts by 20%.
 - Built scalable batch and real-time feature pipelines using Apache Spark, Kafka, Airflow, SQL, Hive, and AWS Glue to process millions of payment transactions, enabling real-time fraud monitoring, low-latency risk scoring, and payment decisioning.
 - Designed customer and merchant behavior models using transaction history, payment attributes, device signals, velocity patterns, and behavioral analytics to support merchant onboarding, transaction anomaly detection, and risk-based decisioning.
 - Developed NLP-based automation solutions using spaCy and transformer models for KYC document classification, merchant onboarding review, customer support ticket routing, and payment dispute categorization, reducing manual review effort by 30%.
 - Deployed production ML models using AWS SageMaker, FastAPI/Flask microservices, Docker, MLflow, and CI/CD pipelines, improving deployment efficiency by 75% while enabling scalable model serving, versioning, and continuous monitoring.
 - Built Power BI and Tableau dashboards to monitor fraud trends, payment success rates, transaction risk metrics, model drift, and false-positive patterns while implementing SHAP-based explainability, GDPR compliance, PII protection, and audit-ready AI governance.

Education

Master of Science in Information Systems and Technologies University of North Texas Denton, TX, USA	08/2022 – 05/2024
Bachelor of Technology in Electrical and Electronics Engineering Guru Nanak Institutions Technical Campus, India	08/2017 – 07/2021

Projects

Agentic RAG System for Enterprise Knowledge Management

- Architected a production-grade multi-agent Retrieval-Augmented Generation (RAG) platform using LangChain, LangGraph, GPT-4, FAISS, and OpenAI embeddings, orchestrating specialized agents for enterprise document retrieval, summarization, reasoning, and fact verification across 10K+ documents.
- Implemented hybrid search combining dense embeddings (Sentence-BERT) and sparse retrieval (BM25), improving answer relevance by 45% compared to single method approaches.
- Deployed on AWS (Lambda, S3, DynamoDB) with automated reindexing pipelines, serving 500+ daily queries with 95% user satisfaction and sub-3s response latency.

Intelligent Healthcare Chatbot for Patient Support

- Developed a production-ready conversational AI chatbot using Python, Rasa, Transformer-based NLP models, Retrieval-Augmented Generation (RAG), and FastAPI to automate patient support while delivering context-aware responses and secure healthcare knowledge retrieval.
- Integrated the chatbot with cloud-hosted databases and deployed via Docker containers on AWS, enabling scalable access, secure patient data handling, and real-time analytics for tracking frequently asked questions and improving system performance.

Certifications

- AI Fluent Tech Professional (Calyptus)
- Oracle Cloud Infrastructure AI Foundations (Oracle)
- Internet of Things Trainee (Adobe)
- SQL for Data Science (Coursera)
- Fundamentals of Digital Marketing (Google Digital Garage)
- Excel Certified Specialist (Great Learning)